

5.2 Kernel Ridge Regression

Ridge regression in a feature space, without building the features.

These notes walk through the lecture slides. The original deck is unchanged; use **(slides)** on the course hub if you want that PDF.

The figures follow Deisenroth, Faisal, and Ong, *Mathematics for Machine Learning*, Theodoridis, and Géron, *Hands-On Machine Learning*.

1. Kernel ridge regression

Regularized empirical risk in an RKHS is

$$\min_{f \in \mathbb{H}} J(f) = \sum_{n=1}^N \mathcal{L}(y_n, f(\mathbf{x}_n)) + \lambda \Omega(\|f\|^2).$$

The regularizer $\Omega(\|f\|^2)$ is typically the squared RKHS norm $\|f\|^2$.

2. Matrix notation

- $\mathbf{y}$: targets.
- $\mathbf{X}$: design matrix of inputs.
- $\mathbf{K}$: **kernel matrix**.

The regularizer becomes $\lambda \theta^\top \mathbf{K} \theta$, where θ holds the coefficients in the RKHS expansion of f .

3. Gradient

The finite-dimensional problem is

$$\min_{\theta} J(\theta) = (\mathbf{y} - \mathbf{K}\theta)^\top (\mathbf{y} - \mathbf{K}\theta) + \lambda \theta^\top \mathbf{K} \theta.$$

Differentiate and set to zero:

$$\frac{\partial J(\theta)}{\partial \theta} = -2\mathbf{K}^\top (\mathbf{y} - \mathbf{K}\theta) + 2\lambda \mathbf{K} \theta = 0.$$

4. Solving for θ

$$\mathbf{K}^\top \mathbf{K} \theta - \mathbf{K}^\top \mathbf{y} + \lambda \mathbf{K} \theta = 0,$$

so

$$(\mathbf{K}^\top \mathbf{K} + \lambda \mathbf{I})\theta = \mathbf{K}^\top \mathbf{y},$$

and

$$\theta = (\mathbf{K}^\top \mathbf{K} + \lambda \mathbf{I})^{-1} \mathbf{K}^\top \mathbf{y}.$$

When $\mathbf{K}$ can be cancelled, the usual kernel-ridge form is

$$\theta = (\mathbf{K} + \lambda \mathbf{I})^{-1} \mathbf{y}.$$

5. Predictions

With that θ ,

$$\hat{\mathbf{y}} = \mathbf{K}\theta.$$

The linear algebra is the same as ridge, but every inner product among features has been replaced by a kernel evaluation. The penalty still controls complexity and **overfitting**.

6. Practice

1. What does the kernel trick avoid computing?
2. Write the usual closed-form θ for kernel ridge in terms of $\mathbf{K}$, λ , and $\mathbf{y}$.
3. Once you have θ , how do you form the vector of training predictions?